Band-Limitedness of $Z(t)$ in the θ-Variable

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Abstract

Let $Z : \mathbb{R} \to \mathbb{R}$ denote Hardy's function, defined by $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)} Z(t)$, where $\theta(t) = \Im \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$ is the Riemann–Siegel theta function. The truncated transform $K_T(\nu) := \frac{1}{2\pi} \int_0^T \zeta(\frac{1}{2} + it)\sqrt{\theta'(t)}\, e^{-i\nu\theta(t)}\, dt$ is shown to satisfy $\lim_{T\to\infty} K_T(\nu) = 0$ pointwise for every $\nu \in \mathbb{R} \setminus [-2, 0]$. The proof rests on exact algebraic cancellation of $\theta''(t(x))$ in the integration-by-parts quotient under the substitution $x = \theta'(t)$, reducing each Riemann–Siegel mode integral to a rational kernel whose decay is controlled to $O(\theta'(T)^{-1/2})$.

1 Introduction

Throughout this paper, ζ denotes the Riemann zeta function and Z denotes Hardy's function, related by $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)} Z(t)$ with $\theta(t) = \Im \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$. Fix $T_0 > 0$ such that $\theta'(T_0) > 0$ and $N(T_0) \geq 1$, where $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$.

Definition 1. *For $T \geq T_0$ and $\nu \in \mathbb{R}$, define the truncated θ-transform of Z by*

$$K_T(\nu) := \frac{1}{2\pi} \int_{T_0}^{T} Z(t)\sqrt{\theta'(t)}\, e^{-i(\nu+1)\theta(t)}\, dt. \tag{1}$$

The substitution $\zeta(\frac{1}{2}+it) = e^{-i\theta(t)} Z(t)$ identifies this with $\frac{1}{2\pi} \int_{T_0}^{T} \zeta(\frac{1}{2}+it)\sqrt{\theta'(t)}\, e^{-i\nu\theta(t)}\, dt$. Set $\mu := \nu + 1$, so that the condition $\nu \notin [-2, 0]$ is equivalent to $|\mu| > 1$.

The principal result stated below establishes pointwise vanishing of $K_T(\nu)$ as $T \to \infty$ for every $\nu \in \mathbb{R} \setminus [-2, 0]$.

2 Statement and Proof

Theorem 1. *For every fixed $\nu \in \mathbb{R} \setminus [-2, 0]$,*

$$\lim_{T\to\infty} K_T(\nu) = 0. \tag{2}$$

Proof. The digamma series $\psi'(s) = \sum_{k\geq 0}(s+k)^{-2}$ evaluated at $s = \frac{1}{4} + \frac{it}{2}$ yields

$$\theta''(t) = \frac{1}{4}\sum_{k\geq 0} \frac{2(\frac{1}{4}+k)\cdot \frac{t}{2}}{\left|\frac{1}{4}+k+\frac{it}{2}\right|^4} > 0 \qquad (t>0). \tag{3}$$

Each summand is strictly positive for $t > 0$, hence $\theta' : [T_0, \infty) \to [\theta'(T_0), \infty)$ is a strictly increasing C^∞ bijection with C^∞ inverse $t : [\theta'(T_0), \infty) \to [T_0, \infty)$, $x \mapsto t(x)$. Moreover $\theta'(t) \to \infty$ as $t \to \infty$ via $\theta'(t) = \frac{1}{2}\log(t/(2\pi)) + O(t^{-2})$.

The Riemann–Siegel identity provides the exact decomposition $Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\theta(t) - t\log n) + R(t)$, where $R(t) = O(t^{-1/4})$ is the Riemann–Siegel remainder. Substituting $\cos\alpha = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha})$ and exchanging the finite sum with the integral yields

$$K_T(\nu) = \frac{1}{2\pi}\sum_{\sigma\in\{+1,-1\}}\sum_{n\leq N(T)} n^{-1/2}\, J_{n,\sigma}(T,\mu) + \mathcal{R}(T,\mu), \tag{4}$$

where, with $t_n := \max(T_0, 2\pi n^2)$,

$$J_{n,\sigma}(T,\mu) := \int_{t_n}^{T} \sqrt{\theta'(t)}\, e^{i[(\sigma-\mu)\theta(t) - \sigma t\log n]}\, dt, \tag{5}$$

and $\mathcal{R}(T,\mu) := \frac{1}{2\pi}\int_{T_0}^{T} R(t)\sqrt{\theta'(t)}\, e^{-i\mu\theta(t)}\, dt$. Since $R(t)\sqrt{\theta'(t)} = O(t^{-1/4}(\log t)^{1/2})$, integration by parts in $\theta(t)$ against the nonstationary phase $e^{-i\mu\theta(t)}$ (the phase derivative is $-\mu\theta'(t) \to -\infty$, hence bounded away from zero on $[T_0,\infty)$) shows $\mathcal{R}(T,\mu)$ converges absolutely as $T\to\infty$ and the corresponding tail $\mathcal{R}(\infty,\mu) - \mathcal{R}(T,\mu)$ is $O(T^{-1/4}(\log T)^{-1/2}) \to 0$.

Set $X := \theta'(T)$, $\beta_n := \theta'(2\pi n^2)$, and $x_0 := \theta'(T_0)$. The substitution $x = \theta'(t)$, $dx = \theta''(t)\, dt$, transforms each mode integral to

$$J_{n,\sigma}(T,\mu) = \int_{\beta_n \vee x_0}^{X} \frac{\sqrt{x}}{\theta''(t(x))}\, e^{i\widetilde{\Phi}_{n,\sigma,\mu}(x)}\, dx, \tag{6}$$

where the phase $\widetilde{\Phi}_{n,\sigma,\mu}(x) := (\sigma-\mu)\theta(t(x)) - \sigma t(x)\log n$ has derivative

$$\widetilde{\Phi}'_{n,\sigma,\mu}(x) = \frac{(\sigma-\mu)x - \sigma\log n}{\theta''(t(x))}. \tag{7}$$

The numerator is affine in x with unique zero $x^*_{n,\sigma} = \sigma\log n/(\sigma-\mu)$. The amplitude-to-phase-derivative ratio admits exact cancellation of $\theta''(t(x))$:

$$\frac{A_n(x)}{\widetilde{\Phi}'_{n,\sigma,\mu}(x)} = \frac{\sqrt{x}/\theta''(t(x))}{[(\sigma-\mu)x - \sigma\log n]/\theta''(t(x))} = \frac{\sqrt{x}}{(\sigma-\mu)x - \sigma\log n} =: Q_{n,\sigma,\mu}(x), \tag{8}$$

a function of x alone. Its derivative is

$$Q'_{n,\sigma,\mu}(x) = -\frac{(\sigma-\mu)x + \sigma\log n}{2\sqrt{x}\,[(\sigma-\mu)x - \sigma\log n]^2}. \tag{9}$$

For $|\mu| > 1$, the unique sign-condition zero $x^*_{n,\sigma} > 0$ occurs in exactly one of the two branches $\sigma \in \{+1,-1\}$, and equals $\log n/(1+|\mu|)$. Define

$$S(\mu) := \left\{n\in\mathbb{N} : \beta_n \leq \tfrac{\log n}{1+|\mu|}\right\}. \tag{10}$$

Lemma 1. *$S(\mu)$ has finite cardinality for every $|\mu| > 1$.*

Proof. The asymptotic $\psi(z) \sim \log z$ implies $\theta'(t) = \frac{1}{2}\log(t/(2\pi)) + O(t^{-2})$, hence $\beta_n = \theta'(2\pi n^2) = \log n + O(n^{-4})$. Therefore

$$\beta_n - \tfrac{\log n}{1+|\mu|} = \tfrac{|\mu|}{1+|\mu|}\log n + O(n^{-4}) \longrightarrow +\infty. \tag{11}$$

There exists $N_0(\mu) < \infty$ such that $n > N_0(\mu) \Rightarrow n \notin S(\mu)$, so $|S(\mu)| \le N_0(\mu) < \infty$. $\square$

For $n \notin S(\mu)$, the zero $x^*_{n,\sigma}$ lies strictly below $\beta_n \vee x_0$, hence $\widetilde{\Phi}'_{n,\sigma,\mu}$ has constant sign on $[\beta_n \vee x_0, X]$ for every $X > \beta_n \vee x_0$, and the denominator $(\sigma-\mu)x - \sigma\log n$ is bounded away from zero. Integration by parts with $u = Q_{n,\sigma,\mu}(x)$, $dv = i\widetilde{\Phi}'_{n,\sigma,\mu}(x)\, e^{i\widetilde{\Phi}_{n,\sigma,\mu}(x)}\, dx$ yields

$$\begin{aligned} J_{n,\sigma}(T,\mu) = {} & \frac{\sqrt{X}}{i\,[(\sigma-\mu)X - \sigma\log n]}\, e^{i\widetilde{\Phi}_{n,\sigma,\mu}(X)} \\ & - \frac{\sqrt{\beta_n \vee x_0}}{i\,[(\sigma-\mu)(\beta_n \vee x_0) - \sigma\log n]}\, e^{i\widetilde{\Phi}_{n,\sigma,\mu}(\beta_n\vee x_0)} \\ & + \int_{\beta_n\vee x_0}^{X} \frac{(\sigma-\mu)x + \sigma\log n}{2i\sqrt{x}\,[(\sigma-\mu)x - \sigma\log n]^2}\, e^{i\widetilde{\Phi}_{n,\sigma,\mu}(x)}\, dx, \end{aligned} \tag{12}$$

which is the exact identity. The boundary at X satisfies

$$\left|\frac{\sqrt{X}}{(\sigma-\mu)X - \sigma\log n}\right| \le \frac{\sqrt{X}}{(|\mu|-1)X - |\log n|} \sim \frac{1}{(|\mu|-1)\sqrt{X}} \longrightarrow 0 \tag{13}$$

as $X \to \infty$. The integrand of the remainder satisfies the absolute bound

$$\left|\frac{(\sigma-\mu)x + \sigma\log n}{2\sqrt{x}\,[(\sigma-\mu)x - \sigma\log n]^2}\right| = O(x^{-3/2}) \qquad (x \to \infty), \tag{14}$$

so the remainder integral converges absolutely on $[\beta_n \vee x_0, \infty)$ and the tail $\int_X^\infty |\cdot|\, dx = O(X^{-1/2}) \to 0$.

For $n \in S(\mu)$, the stationary point $t^*_{n,\sigma} = t(x^*_{n,\sigma})$ is a fixed finite value determined by $\theta'(t^*_{n,\sigma}) = \log n/(1+|\mu|)$. The mode $J_{n,\sigma}(T,\mu)$ converges as $T \to \infty$ to a finite limit $J_{n,\sigma}(\infty,\mu) \in \mathbb{C}$, contributing to $K_\infty(\nu)$. The quantity to control is the tail

$$J_{n,\sigma}(\infty,\mu) - J_{n,\sigma}(T,\mu) = \int_T^\infty \sqrt{\theta'(t)}\, e^{i\Phi_{n,\sigma,\mu}(t)}\, dt, \tag{15}$$

where $\Phi_{n,\sigma,\mu}(t) = (\sigma-\mu)\theta(t) - \sigma t\log n$. For $T > t^*_{n,\sigma}$, strict monotonicity of θ' gives $\theta'(T) > \log n/(1+|\mu|)$, whence

$$|\Phi'_{n,\sigma,\mu}(t)| = |(\sigma-\mu)\theta'(t) - \sigma\log n| \ge (|\mu|-1)\big(\theta'(T) - \tfrac{\log n}{|\mu|-1}\big) > 0 \tag{16}$$

for all $t \ge T$, with the lower bound diverging as $T \to \infty$. The denominator in the integration-by-parts quotient is therefore bounded away from zero on $[T,\infty)$, and the argument applied to $n \notin S(\mu)$ transfers verbatim to the tail integral, yielding upper boundary $O(\theta'(T)^{-1/2})$ and remainder $O(\theta'(T)^{-1/2})$, both vanishing as $T \to \infty$.

The cardinality $|S(\mu)|$ is finite, so the aggregate contribution from $\{n \in S(\mu)\}$ is a finite sum of terms each tending to zero. Combined with the vanishing of $\mathcal{R}(\infty, \mu) - \mathcal{R}(T, \mu)$ and of the $n \notin S(\mu)$ tails, every component of $K_\infty(\nu) - K_T(\nu)$ tends to zero as $T \to \infty$. Therefore

$$\lim_{T \to \infty} K_T(\nu) = 0 \qquad \text{for every fixed } \nu \in \mathbb{R} \setminus [-2, 0]. \tag{17}$$

$\square$